

on the basis of application and technology.

The application segment comprises food toxicity detection, industrial process control, medicine/diagnostics/point-of-care testing and other application areas such as environment and agriculture. The industry is witnessing increasing use of these devices in laboratories, point-of-care testing and self-testing.

Medical and diagnostics applications including POC testing dominate the market. POC applications can be segmented into glucose monitoring, cardiac markers, infectious disease diagnosis, coagulation monitoring, pregnancy and fertility testing, blood gas and electrolytes testing, tumour or cancers markers, urinalysis testing, cholesterol tests, etc. Cardiac markers are the most dynamic POC application and are expected to witness several innovations and investments over the next five years, owing to the increasing number of people suffering from cardiovascular diseases and rising demand for instant diagnosis of these diseases.

By 2022, glucose monitoring is likely to hold the largest share of biosensors market for point-of-care applications. Blood glucose monitoring plays a crucial role in diabetes management. Glucose biosensors provide real-time information on the changes in glucose concentration.

BIOSENSOR MARKET SEGMENTATION

By application	By technology type
- Food toxicity/allergen detection - Industrial process control - Medicine/diagnostics/healthcare/point-of-care testing - Environment - Agriculture - Security and bio-defence	- Electrochemical - Thermal - Optical - Piezoelectric - Nanomechanical - Others

The glucose biosensors technology helps maintain normal blood glucose levels. The rising prevalence of diabetes worldwide, technological advancements in self-monitoring of blood glucose, and growing awareness about POC testing are the key factors boosting the demand for glucose monitoring biosensors. Among all applications, glucose monitoring held the largest size of biosensors in 2016.

The glucose biosensors industry is segmented on the basis of its end use in hospitals, homecare diagnostics and others including research institutes and diagnostics and clinical laboratories. The homecare diagnostics market for glucose biosensors is projected to flourish over the forecast period 2015-22. Ease-of-use, quick results and low costs of the tests are supporting market growth. On the other hand, hospital-based market is projected to grow at a significant CAGR of 10.3 per cent during this period.

Are the Indian transducer suppliers ready to capture biosensor market?

Forty per cent of the survey participants felt that the biosensor sector can open up business opportunities for them (Fig. 2). However, at present, none of them is supplying transducers to the biosensor industry.

A few participants see applications of biosensors beyond medical or point-of-care testing because of the demand from agriculture sector.



Fig. 2: Business opportunities for transducers in biosensors sector

Based on the technology, the biosensors market is segmented into electrochemical, thermal, optical and piezoelectric markets.

Innovations in electrochemical sensors with high-throughput methods focusing on detection limit, analysis time and portability have provided large-scale consumer markets for inexpensive biosensors in glucose and pregnancy tests.

Optic-based biosensors involving fibre-optic chemistry are the next major technology in biosensing. Optical biosensors for DNA measurement find applications in biomedicine and forensic science.

Moreover, integrated strategies using multiple technologies ranging from electrochemical, electromechanical and fluorescence-cum-optical-based biosensors and genetically engineered microbes are modern methods for biosensor discoveries. Development of different micro- and nano-biosensor platforms involving integrated technologies utilising electrochemical or optic bio-electronic principles with a combination of biomolecules or biological materials, polymers and nanomaterials is required to achieve highly sensitive and miniaturised devices. **EFY**

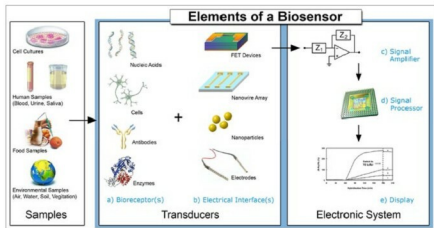


Fig. 3: Elements of a biosensor

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